

Lesson 32: Area of Polar Region

Criteria	4 - Advanced	3 - Proficient	2 - Developing	1 - Beginning
Setup & Limits (30%)	Correctly identifies intersection points and sets up the integral with precise limits.	Identifies most intersection points; sets up integral with minor errors in limits.	Incorrect intersection points, but correct formulaic structure.	Incorrect formula ($\frac{1}{2}r^2$) and incorrect limits.
Integrand (30%)	Correctly identifies $r(\theta)$ and sets up the full integrand: $\frac{1}{2}r^2 d\theta$ or $\frac{1}{2}(r_{out}^2 - r_{in}^2)d\theta$.	Correct formula structure with minor errors in substituting $r(\theta)$.	Misidentifies the inner/outer curve or squares incorrectly.	Missing $\frac{1}{2}$ or or fails to square $r(\theta)$.
Integration (20%)	Accurately evaluates the definite integral, either analytically or with technology.	Evaluates with minor errors in substitution or power reduction formulas.	Struggles with integration techniques, leading to significant errors.	Cannot set up or compute the integral.
Final Answer (20%)	Final answer is numerically correct, rounded to 3 decimal places.	Final answer is close but lacks required precision.	Final answer is incorrect but shows a logical path.	Answer is missing or irrelevant.